Adult—Respiratory

CHAPTER 51

Respiratory Problems

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PRIORITY CONCEPTS Gas Exchange; Perfusion

I. Anatomy and Physiology

A. Primary functions of the respiratory system
  1. Provides oxygen for metabolism in the tissues
  2. Removes carbon dioxide, the waste product of metabolism

B. Secondary functions of the respiratory system
  1. Facilitates sense of smell
  2. Produces speech
  3. Maintains acid–base balance
  4. Maintains body water levels
  5. Maintains heat balance

C. Upper respiratory airway
  1. Nose: Humidifies, warms, and filters inspired air
  2. Sinuses: Air-filled cavities within the hollow bones that surround the nasal passages and provide resonance during speech
  3. Pharynx
    a. Passageway for the respiratory and digestive tracts located behind the oral and nasal cavities
    b. Divided into the nasopharynx, oropharynx, and laryngopharynx
  4. Larynx
    a. Located just below the pharynx at the root of the tongue; commonly called the *voice box*
    b. Contains two pairs of vocal cords, the false and true cords
    c. The opening between the true vocal cords is the glottis. The glottis plays an important role in coughing, which is the most fundamental defense mechanism of the lungs.
  5. Epiglottis
    a. Leaf-shaped elastic flap structure at the top of the larynx
    b. Prevents food from entering the tracheobronchial tree by closing over the glottis during swallowing

D. Lower respiratory airway
  1. Trachea: Located in front of the esophagus; branches into the right and left mainstem bronchi at the carina
  2. Mainstem bronchi
    a. Begin at the carina
    b. The right bronchus is slightly wider, shorter, and more vertical than the left bronchus.
    c. Divide into secondary or lobar bronchi that enter each of the five lobes of the lung
    d. The bronchi are lined with cilia, which propel mucus up and away from the lower airway to the trachea, where it can be expectorated or swallowed.
  3. Bronchioles
    a. Branch from the secondary bronchi and subdivide into the small terminal and respiratory bronchioles
    b. Contain no cartilage and depend on the elastic recoil of the lung for patency
    c. The terminal bronchioles contain no cilia and do not participate in gas exchange.
  4. Alveolar ducts and alveoli
    a. *Acinus* (plural, *acini*) is a term used to indicate all structures distal to the terminal bronchiole.
    b. Branch from the respiratory bronchioles
    c. Alveolar sacs, which arise from the ducts, contain clusters of alveoli, which are the basic units of gas exchange.
    d. Type 2 alveolar cells in the walls of the alveoli secrete surfactant, a phospholipid protein that reduces the surface tension in the alveoli; without surfactant, the alveoli would collapse.
  5. Lungs
    a. Located in the pleural cavity in the thorax
    b. Extend from just above the clavicles to the diaphragm, the major muscle of inspiration

c. The right lung, which is larger than the left, is divided into three lobes: the upper, middle, and lower lobes.
d. The left lung, which is narrower than the right lung to accommodate the heart, is divided into two lobes.
e. The respiratory structures are innervated by the phrenic nerve, the vagus nerve, and the thoracic nerves.
f. The parietal pleura lines the inside of the thoracic cavity, including the upper surface of the diaphragm.
g. The visceral pleura covers the pulmonary surfaces.
h. A thin fluid layer, which is produced by the cells lining the pleura, lubricates the visceral pleura and the parietal pleura, allowing them to glide smoothly and painlessly during respiration.
i. **Blood** flows throughout the lungs via the pulmonary circulation system.

6. Accessory muscles of respiration include the scalene muscles, which elevate the first two ribs; the sternocleidomastoid muscles, which raise the sternum; and the trapezius and pectoralis muscles, which fix the shoulders.
7. The respiratory process
   a. The diaphragm descends into the abdominal cavity during inspiration, causing negative pressure in the lungs.
   b. The negative pressure draws air from the area of greater pressure, the atmosphere, into the area of lesser pressure, the lungs.
   c. In the lungs, air passes through the terminal bronchioles into the alveoli and diffuses into surrounding capillaries, then travels to the rest of the body to oxygenate the body tissues.
   d. At the end of inspiration, the diaphragm and intercostal muscles relax and the lungs recoil.
   e. As the lungs recoil, pressure within the lungs becomes higher than atmospheric pressure, causing the air, which now contains the cellular waste products carbon dioxide and water, to move from the alveoli in the lungs to the atmosphere.
   f. Effective gas exchange depends on distribution of gas (ventilation) and blood (perfusion) in all portions of the lungs.

II. Diagnostic Tests

A. Risk factors for respiratory disorders (Box 51.1)

B. Chest x-ray film (radiograph)
1. Description: Provides information regarding the anatomical location and appearance of the lungs
2. Preprocedure
   a. Remove all jewelry and other metal objects from the chest area.

BOX 51.1 Risk Factors for Respiratory Disorders

- Chest injury
- Crowded living conditions
- Environmental allergies
- Exposure to chemicals and environmental pollutants
- Family history of infectious disease
- Frequent respiratory illnesses
- Geographical residence and travel to foreign countries
- Smoking
- Surgery
- Use of chewing tobacco
- Viral syndromes

Reference: Ignatavicius, D., Workman, M., Rebar, C., & Heimgartner, N. (2021). *Concepts for interprofessional collaborative care*. (10th ed.). St. Louis: Saunders.

   b. Assess the client's ability to inhale and hold their breath.
3. Postprocedure: No special care is required after the procedure unless there are abnormal findings.

Question the client regarding pregnancy or the possibility of pregnancy before performing radiography studies.

C. Sputum specimen
1. Description: Specimen obtained by expectoration or tracheal **suctioning** to assist in the identification of organisms or abnormal cells (see Box 70.11 in Chapter 70)
2. Sputum for culture and sensitivity should be collected before antimicrobial therapy is initiated unless the test is being performed to evaluate the effectiveness of medications already being given.
3. Preprocedure
   a. Determine the specific purpose of collection, and check institutional policy for the appropriate method for collection.
   b. Obtain an early morning sterile specimen by suctioning or expectoration after a respiratory treatment if a treatment is prescribed; give the client the specimen cup the night before.
   c. Instruct the client to rinse the mouth with water before collection to decrease contamination of the sputum sample from particles in the oropharynx.
   d. Obtain 15 mL of sputum.
   e. Instruct the client to take several deep breaths and then cough deeply to obtain sputum. Remind the client that sputum comes from the lungs and that saliva is not sputum.
   f. Collect the specimen before the client begins antibiotic therapy. If already started on antibiotic therapy, ensure that the laboratory can utilize an antimicrobial removal device when analyzing the specimen.

4. Postprocedure
   a. If a culture of sputum is prescribed, transport the labeled specimen to the laboratory immediately; indicate whether the client was currently on antimicrobial therapy at the time of collection.
   b. Assist the client with mouth care.

Ensure that an informed consent was obtained for any invasive procedure. Vital signs are measured before the procedure and monitored postprocedure to detect signs of complications.

D. Laryngoscopy and bronchoscopy
1. Description: Direct visual examination of the larynx, trachea, and bronchi with a fiberoptic bronchoscope
2. Preprocedure
   a. Maintain NPO (nothing by mouth) status as prescribed.
   b. Assess the results of coagulation studies.
   c. Remove dentures and eyeglasses.
   d. Instruct the client to perform good mouth care to prevent bacteria from entering into the lungs from the oropharynx.
   e. Establish an intravenous (IV) access as necessary, and administer medication for sedation as prescribed. A local anesthetic spray may be used; instruct the client not to swallow the spray and to expectorate any excess into a basin.
   f. Have emergency resuscitation supplies readily available.
3. Postprocedure
   a. Maintain the client in a semi-Fowler's position.
   b. Assess for the return of the gag reflex.
   c. Maintain NPO status until the gag reflex returns.
   d. Monitor for bloody sputum.
   e. Monitor respiratory status, particularly if sedation has been administered.
   f. Monitor for complications, such as bronchospasm or bronchial perforation, indicated by facial or neck crepitus, dysrhythmias, hemorrhage, hypoxemia, and **pneumothorax.**
   g. Notify the primary health care provider (PHCP) if signs of complications occur.
   h. Inform the client that warm saline gargles and lozenges may be helpful for a sore throat.
   i. Biopsy or culture results are available in 2 to 7 days.

E. Endobronchial ultrasound (EBUS)
1. Tissue samples are obtained from central lung masses and lymph nodes, using a bronchoscope with the help of ultrasound guidance.
2. Tissue samples are used for diagnosing and staging lung cancer, detecting infections, and identifying inflammatory diseases that affect the lungs, such as sarcoidosis.
3. Postprocedure, the client is monitored for signs of bleeding and respiratory distress.

F. Pulmonary angiography
1. Description
   a. A fluoroscopic procedure in which a catheter is inserted through the antecubital or femoral vein into the pulmonary artery or one of its branches
   b. Involves an injection of iodine or radiopaque contrast material
2. Preprocedure
   a. Assess for allergies to iodine, seafood, or other radiopaque dyes.
   b. Maintain NPO status as prescribed.
   c. Assess results of coagulation studies.
   d. Establish an IV access.
   e. Administer sedation as prescribed.
   f. Instruct the client to lie still during the procedure.
   g. Instruct the client that they may feel an urge to cough, flushing, nausea, or a salty taste following injection of the dye.
   h. Have emergency resuscitation equipment available.
3. Postprocedure
   a. Avoid taking **blood pressures** for 24 hours in the extremity used for the injection.
   b. Monitor peripheral neurovascular status of the affected extremity.
   c. Assess insertion site for bleeding.
   d. Monitor for reaction to the dye.
   e. Apply cold compresses to the puncture site to reduce swelling or discomfort.

G. Thoracentesis
1. Description: Removal of fluid or air from the pleural space via transthoracic aspiration
2. Preprocedure
   a. Prepare the client for ultrasound or chest radiograph, if prescribed, before the procedure.
   b. Assess results of coagulation studies.
   c. Note that the client is positioned sitting upright, with the arms and shoulders supported by a table at the bedside during the procedure (Fig. 51.1).
   d. If the client cannot sit up, the client is placed lying in bed toward the unaffected side, with the head of the bed elevated.
   e. Instruct the client not to cough, breathe deeply, or move during the procedure.
   f. Administer cough suppressant as prescribed before the procedure if the client has a troublesome cough.

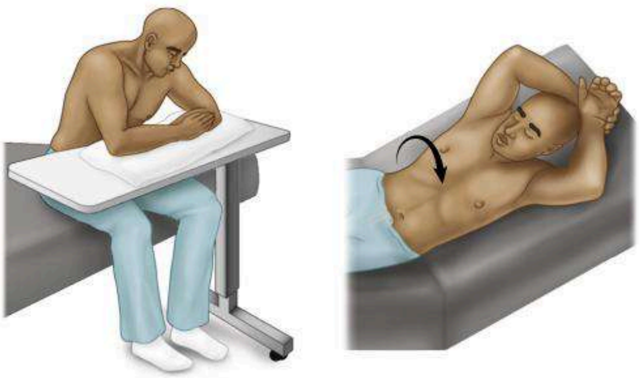

FIG. 51.1 Positions for thoracentesis.

3. Postprocedure
   a. Monitor respiratory status.
   b. Apply a pressure dressing, and assess the puncture site for bleeding and crepitus.
   c. Monitor for signs of pneumothorax, air embolism, and pulmonary edema.
   d. Review chest x-ray results to monitor for pneumothorax.

H. Pulmonary function tests
1. Description: Tests used to evaluate lung mechanics, gas exchange, and acid–base disturbance through spirometric measurements, lung volumes, and arterial blood gas levels
2. Preprocedure
   a. Determine whether an analgesic that may depress the respiratory function is being administered.
   b. Consult with the PHCP regarding withholding bronchodilators before testing, or alternatively whether the testing will be done prior to and after administration of a bronchodilator.
   c. Instruct the client to void before the procedure and to wear loose clothing.
   d. Remove dentures.
   e. Instruct the client to refrain from smoking or eating a heavy meal for 4 to 6 hours before the test.
   f. Measure height and weight to determine predictive values.
3. Postprocedure: The client may resume a normal diet and any bronchodilators and respiratory treatments that were withheld before the procedure.

Clients with severe respiratory problems are occasionally exhausted after the testing and will need rest.

I. Lung biopsy
1. Description
   a. A transbronchial biopsy and a transbronchial needle aspiration may be performed to obtain tissue for analysis by culture or cytological examination.
   b. An open lung biopsy is performed in the operating room.
2. Preprocedure
   a. Maintain NPO status as prescribed.
   b. Inform the client that a local anesthetic will be used for a needle biopsy but that a sensation of pressure during needle insertion and aspiration may be felt.
   c. Administer analgesics and sedatives as prescribed.
   d. Instruct client to remain still during the procedure.
3. Postprocedure
   a. Apply a dressing to the biopsy site and monitor for drainage or bleeding.
   b. Monitor for signs of respiratory distress, and notify the PHCP if they occur.
   c. Monitor for signs of pneumothorax and air emboli, and notify the PHCP if they occur.
   d. Prepare the client for chest radiography if prescribed.

J. Spiral (helical) computed tomography (CT) scan
1. Frequently used test to diagnose pulmonary embolism
2. IV injection of contrast medium is used; if the client cannot have contrast medium, a ventilation-perfusion (V/Q) scan will be done.
3. The scanner rotates around the body, allowing for a three-dimensional picture of all regions of the lungs.

K. V/Q lung scan
1. Description
   a. The perfusion scan evaluates blood flow to the lungs.
   b. The ventilation scan determines the patency of the pulmonary airways and detects abnormalities in ventilation.
   c. A radionuclide may be injected for the procedure.
   d. Generally this is the preferred test to use with renal impairment.
   e. Encourage the client to drink fluids to avoid renal impairment.
2. Preprocedure
   a. Assess the client for allergies to dye, iodine, or seafood.
   b. Remove jewelry around the chest area.
   c. Review breathing methods that may be required during testing.
   d. Establish an IV access.
   e. Administer sedation if prescribed.
   f. Have emergency resuscitation equipment available.
3. Postprocedure
   a. Monitor the client for reaction to the radionuclide.

b. Instruct the client that the radionuclide clears from the body in about 8 hours.
c. Encourage increased fluid intake to clear the dye from the body if there is no fluid restriction.

L. Computed tomography pulmonary angiography
1. Description
a. The scan visualizes the pulmonary arteries and blood flow.
b. Its main use is to diagnose pulmonary embolism and is the preferred method.
c. A contrast dye is injected.
2. Preprocedure: Similar to the V/Q lung scan; in addition, renal function should be adequate, and dosing of the contrast should be done by a pharmacist.
3. Postprocedure: Similar to the V/Q lung scan

M. Skin tests: A skin test uses an intradermal injection to help diagnose various infectious diseases (Box 51.2).

N. Arterial blood gases (ABGs)
1. Description: Measurement of the dissolved oxygen and carbon dioxide in the arterial blood helps indicate the acid–base state and how well oxygen is being carried to the body.
2. Preprocedure and postprocedure care, normal results, and analysis of results: See Chapter 9.

Avoid suctioning the client before drawing an ABG sample, because the suctioning procedure will deplete the client's oxygen, resulting in inaccurate ABG results.

O. Pulse oximetry: See Chapter 10.

P. D-dimer
1. A blood test that measures clot formation and lysis that results from the degradation of fibrin
2. Helps diagnose (a positive test result) the presence of thrombus in conditions such as deep vein thrombosis, pulmonary embolism, or stroke; it is also used to diagnose disseminated intravascular coagulation (DIC) and to monitor the effectiveness of treatment.
3. D-dimer has high sensitivity, low specificity for diagnosing clot formation.
4. The normal D-dimer level is less than 50 ng/mL (less than 3.0 mmol/L); normal fibrinogen is 60 to 100 mg/dL (2.0 to 5.0 g/L).

III. Respiratory Treatments (see Chapter 70)

IV. Chest Injuries

A. Rib fracture
1. Description
a. Results from direct blunt chest trauma and causes a potential for intrathoracic injury, such as pneumothorax, hemothorax, or pulmonary contusion

BOX 51.2 Tuberculin Skin Test Procedure

1. Determine hypersensitivity or previous reactions to skin tests.
2. Assess whether the client has received Bacille-Calmette-Guerin (BCG) in the past, which would demonstrate a positive reaction.
3. Use a skin site that is free of excessive body hair, dermatitis, and blemishes.
4. Apply the injection at the upper third of the inner surface of the left arm.
5. Circle and mark the injection test site with indelible ink.
6. Document the date, time, and test site.
7. Advise the client not to scratch the test site to prevent infection and possible abscess formation.
8. Instruct the client to avoid washing the test site.
9. Assess the reaction at the injection site 48 to 72 hours after administration of the test antigen.
10. Assess the test site for the amount of induration (hard swelling) in millimeters and for the presence of erythema and vesiculation (small blister-like elevations).

b. Pain with movement, deep breathing, and coughing results in impaired ventilation and inadequate clearance of secretions.
c. Can be serious and life-threatening when three or more ribs are fractured, with preexisting disease (particularly cardiopulmonary disease), or for the elderly client
2. Assessment
a. Pain and tenderness at the injury site that increases with inspiration
b. Shallow respirations
c. Client splints chest using a pillow as needed; external splints are not recommended because they limit chest wall expansion.
d. Fractures noted on rib series x-ray
3. Interventions
a. Note that the ribs usually reunite spontaneously.
b. Client has a higher risk of developing pneumonia after rib fractures.
c. Open reduction and internal fixation of the ribs (rib plating) may be done.
d. Place the client in a Fowler's position.
e. Administer pain medication as prescribed to maintain adequate ventilatory status.
f. Monitor for increased respiratory distress.
g. Instruct the client to self-splint with the hands, arms, or a pillow.
h. Prepare the client for an intercostal nerve block as prescribed if the pain is severe.

B. Flail chest
1. Description
a. Occurs from blunt chest trauma associated with accidents, which may result in hemothorax and rib fractures

b. The loose segment of the chest wall becomes paradoxical to the expansion and contraction of the rest of the chest wall.

2. Assessment
 a. Paradoxical respirations (inward movement of a segment of the thorax during inspiration with outward movement during expiration)
 b. Severe pain in the chest
 c. Dyspnea
 d. **Cyanosis**
 e. Tachycardia
 f. Hypotension
 g. Tachypnea, shallow respirations
 h. Diminished breath sounds
3. Interventions
 a. Maintain the client in a Fowler's position if cervical spine injury has been ruled out.
 b. Administer oxygen as prescribed.
 c. Monitor for increased respiratory distress.
 d. Encourage coughing and deep breathing.
 e. Administer pain medication as prescribed.
 f. Maintain bed rest and limit activity to reduce oxygen demands.
 g. Open reduction and internal fixation of the ribs (rib plating) may be done.
 h. Prepare for intubation with **mechanical ventilation**, with positive end-expiratory pressure (PEEP) for severe flail chest associated with respiratory failure and shock (see Chapter 70).

C. Pulmonary contusion
1. Description
 a. Characterized by interstitial hemorrhage associated with intra-alveolar hemorrhage, resulting in decreased pulmonary compliance
 b. The major complication is acute respiratory distress syndrome.
2. Assessment
 a. Dyspnea
 b. Restlessness
 c. Increased bronchial secretions
 d. Hypoxemia
 e. Hemoptysis
 f. Decreased breath sounds
 g. **Crackles** and wheezes
3. Interventions
 a. Maintain a patent airway and adequate ventilation.
 b. Place the client in a Fowler's position.
 c. Administer oxygen as prescribed.
 d. Monitor for increased respiratory distress.
 e. Maintain bed rest and limit activity to reduce oxygen demands.
 f. Provide oxygen and give fluids as prescribed.
 g. Prepare for mechanical ventilation with PEEP if required.

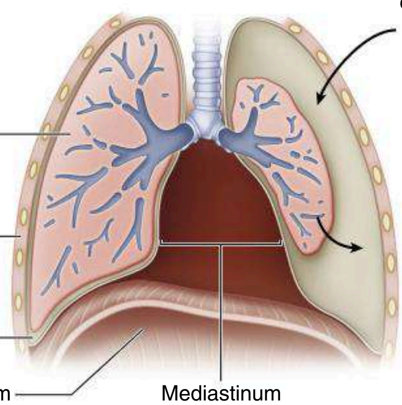

FIG. 51.2 Pneumothorax. Air in the pleural space causes the lungs to collapse around the hilus and may push the mediastinal contents (heart and great vessels) toward the other lung.

D. **Pneumothorax** (Fig. 51.2)
1. Description
 a. Accumulation of atmospheric air in the pleural space, which results in a rise in intrathoracic pressure and reduced vital capacity, or the greatest amount of air expired from the lungs after taking a deep breath
 b. The loss of negative intrapleural pressure results in collapse of the lung.
 c. A spontaneous pneumothorax occurs with the rupture of a pulmonary bleb, or small air-containing spaces deep in the lung.
 d. An open pneumothorax occurs when an opening through the chest wall allows the entrance of positive atmospheric air pressure into the pleural space.
 e. A tension pneumothorax occurs from a blunt chest injury or from mechanical ventilation with PEEP when a buildup of positive pressure occurs in the pleural space.
2. Assessment (Box 51.3)
3. Interventions
 a. Diagnosis of pneumothorax is made by chest x-ray, which will show air or fluid in the pleural space and reduction of lung volume.
 b. Apply a nonporous dressing over an open chest wound.
 c. Administer oxygen as prescribed.
 d. Place the client in a Fowler's position for chest trauma.
 e. Prepare for **chest tube** placement, which will remain in place until the lung has expanded fully.
 f. Monitor the chest tube drainage system.
 g. Monitor for subcutaneous emphysema.
 h. Review serial chest x-ray results to determine effectiveness of treatment.
 i. See Chapter 70 for information on caring for a client with chest tubes.

BOX 51.3 Assessment Findings: Pneumothorax

- Absent or markedly decreased breath sounds on affected side
- Cyanosis
- Decreased chest expansion unilaterally
- Distended neck veins
- Dyspnea
- Hypotension
- Sharp chest pain
- Subcutaneous emphysema as evidenced by crepitus on palpation
- Sucking sound with open chest wound
- Tachycardia
- Tachypnea
- Tracheal deviation to the unaffected side with tension pneumothorax

Clients with a respiratory disorder should be positioned with the head of the bed elevated.

V. Asthma (Fig. 51.3)

A. Description
1. Chronic inflammatory disorder of the airways that causes varying degrees of obstruction in the airways
2. Marked by airway inflammation and hyperresponsiveness to a variety of stimuli or triggers (Box 51.4).
3. Causes recurrent episodes of wheezing, breathlessness, chest tightness, and coughing associated with airflow obstruction that may resolve spontaneously; it is often reversible with treatment.
4. Severity is classified based on the clinical features before treatment.
5. Status asthmaticus is a severe life-threatening **asthma** episode that is refractory to treatment and may result in pneumothorax, acute cor pulmonale, or respiratory arrest.
6. Refer to Chapter 36 for additional information on asthma.

Silent breath sounds are associated with acute asthma exacerbation, may indicate impending respiratory failure due to diffuse bronchospasm, and represent a life-threatening condition.

B. Assessment
1. Restlessness
2. Wheezing or crackles
3. Absent or diminished lung sounds
4. Hyperresonance
5. Use of accessory muscles for breathing
6. Tachypnea with hyperventilation
7. Prolonged exhalation
8. Tachycardia
9. Pulsus paradoxus
10. Diaphoresis
11. Cyanosis
12. Decreased oxygen saturation
13. Pulmonary function test results that demonstrate decreased airflow rates

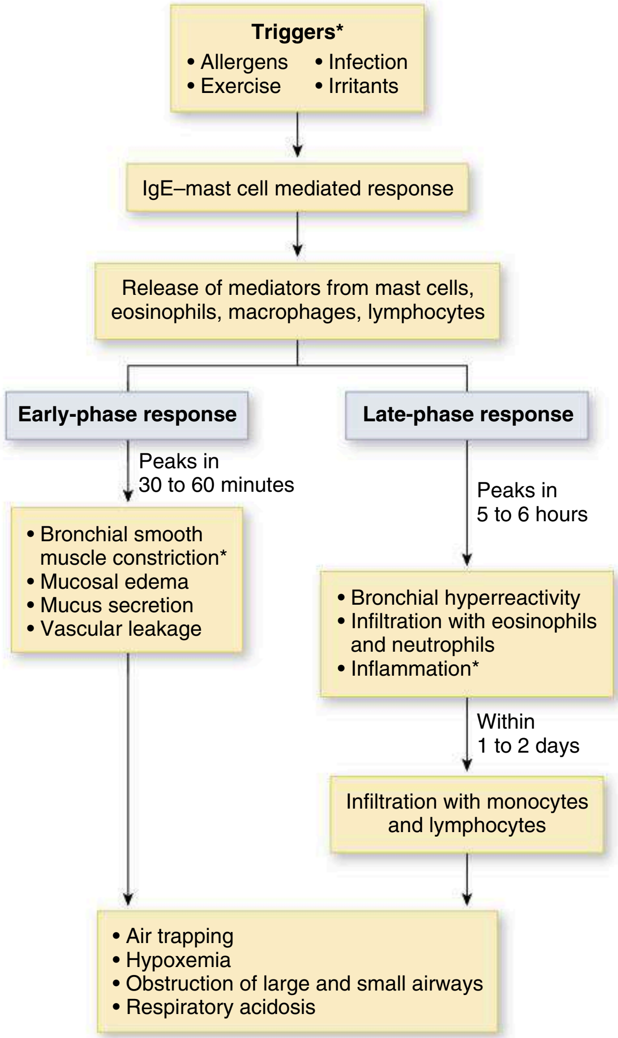

FIG. 51.3 Pathophysiology in asthma. Stems with asterisks are primary processes. *IgE*, Immunoglobulin E.

C. Interventions
1. Monitor vital signs.
2. Monitor pulse oximetry.
3. Monitor peak flow.
4. Administer bronchodilators and corticosteroids as prescribed.
5. Educate the client and family on reducing stress and anxiety and avoiding environmental triggers.
6. During an acute asthma episode, provide interventions to assist with breathing (Box 51.5).

D. Client education
1. Discuss the intermittent nature of symptoms and the need for long-term management.

BOX 51.4 Asthma Triggers

Environmental Factors
- Animal dander
- Cockroaches
- Cold, dry air
- Dust mites
- Exhaust fumes
- Fireplaces
- Molds
- Perfumes or other products with aerosol sprays
- Pollen
- Smoke, including cigarette or cigar smoke
- Sudden weather changes

Physiological Factors
- Exercise
- Gastroesophageal reflux disease (GERD)
- Hormonal changes
- Sinusitis
- Stress
- Viral upper respiratory infection

Medications
- Acetylsalicylic acid (aspirin)
- β-Adrenergic blockers
- Nonsteroidal antiinflammatory drugs

Occupational Exposure Factors
- Agriculture
- Industrial chemicals and plastics
- Metal salts
- Pharmaceutical drugs
- Wood and vegetable dusts

Food Additives
- Beer, wine, dried fruit, shrimp, processed potatoes
- Monosodium glutamate
- Sulfites (bisulfites and metabisulfites)
- Tartrazine

Reference: Lewis, S., Harding, M., Kwong, J., Roberts, D., Hagler, D., & Reinisch, C. (2020). *Medical-surgical nursing: Assessment and management of clinical problems.* (11th ed.). St. Louis: Mosby. pp. 542-543.

2. Identify possible triggers and measures to prevent episodes.
3. Use pursed-lip breathing before, during, and after activities that are possible triggers.
4. Management of medication and proper administration
5. Correct use of a peak flowmeter and aerochamber or "spacer" use with inhaler
6. Wear a MedicAlert bracelet.
7. Develop an asthma action plan with the PHCP and what to do if an asthma episode occurs.

VI. Chronic Obstructive Pulmonary Disease

A. Description
1. Also known as chronic obstructive lung disease and chronic airflow limitation

BOX 51.5 Nursing Interventions During an Acute Asthma Episode

- Position the client in a high-Fowler's position or sitting to aid in breathing.
- Administer oxygen as prescribed.
- Stay with the client to decrease anxiety.
- Administer bronchodilators and other nebulizer treatments as prescribed.
- Record the color, amount, and consistency of sputum, if any.
- Administer corticosteroids as prescribed.
- Administer magnesium sulfate as prescribed.
- Administer intravenous fluids as prescribed.
- Auscultate lung sounds before, during, and after treatments.

2. **Chronic obstructive pulmonary disease** is a disease state characterized by airflow obstruction.
3. Chronic bronchitis and **emphysema** are progressive lung diseases that fall under the general category of chronic obstructive pulmonary disease.
4. Chronic bronchitis is a condition in which the bronchial tubes become inflamed and excessive mucus production occurs as a result of irritants or injury.
5. Emphysema is a condition in which the air sacs in the lungs are damaged and enlarged, resulting in hyperinflation and breathlessness.
6. Progressive airflow limitation occurs, associated with an abnormal inflammatory response of the lungs that is not completely reversible.
7. Chronic obstructive pulmonary disease (COPD) leads to pulmonary insufficiency, pulmonary hypertension, and cor pulmonale.

B. Assessment
1. Cough
2. Exertional dyspnea
3. Wheezing and crackles
4. Sputum production
5. Weight loss
6. Barrel chest (emphysema) (Fig. 51.4)
7. Use of accessory muscles for breathing
8. Prolonged expiration
9. Orthopnea
10. Cyanosis
11. Delayed capillary refill
12. Finger clubbing
13. Cardiac dysrhythmias
14. Congestion and hyperinflation seen on chest x-ray (Fig. 51.5)
15. ABG levels that indicate respiratory acidosis with or without compensation and hypoxemia. Uncompensated respiratory acidosis in a COPD patient generally indicates an acute exacerbation of COPD.

16. Pulmonary function tests that demonstrate decreased vital capacity

C. Interventions
1. Monitor vital signs.
2. Administer a concentration of oxygen based on ABG values and oxygen saturation by pulse oximetry as prescribed.
3. Monitor pulse oximetry.
4. Provide respiratory treatments and chest physiotherapy (CPT).
5. Instruct the client in diaphragmatic or abdominal breathing techniques, tripod positioning, and pursed-lip breathing techniques, which increase airway pressure and keep air passages open, promoting maximal carbon dioxide expiration.
6. Record the color, amount, and consistency of sputum.

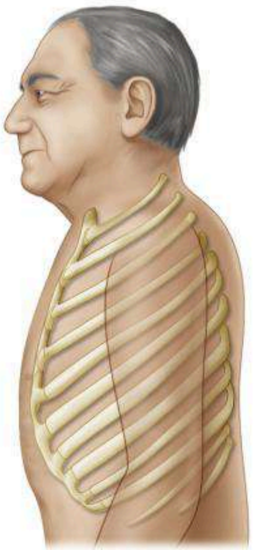

FIG. 51.4 Typical barrel chest in a client with chronic obstructive pulmonary disease.

7. Suction the client's lungs, if necessary, to clear the airway and prevent infection.
8. Monitor weight.
9. Encourage small, frequent meals to maintain nutrition and prevent dyspnea.
10. Provide a high-calorie, high-protein diet with supplements.
11. Encourage fluid intake up to 3000 mL/day to keep secretions thin, unless contraindicated.
12. Place the client in a Fowler's position and leaning forward to aid in breathing (Fig. 51.6).
13. Allow activity as tolerated; include exercise conditioning and pulmonary rehabilitation to prevent muscle deconditioning; assess physical limitations and develop an activity plan based on limitations.
14. Administer bronchodilators and other nebulizer treatments as prescribed, and instruct the client in the use of oral and inhalant medications.
15. Administer corticosteroids as prescribed for exacerbations.
16. Administer mucolytics as prescribed to thin secretions.
17. Administer antibiotics for infection if prescribed.

D. Client education (Box 51.6)

VII. Pneumonia

A. Description
1. Infection of the pulmonary tissue, including the interstitial spaces, the alveoli, and the bronchioles
2. The edema associated with inflammation stiffens the lung, decreases lung compliance and vital capacity, and causes hypoxemia.
3. Pneumonia can be community-acquired or hospital-acquired.
4. The chest x-ray film shows lobar or segmental consolidation, or pulmonary infiltrates.

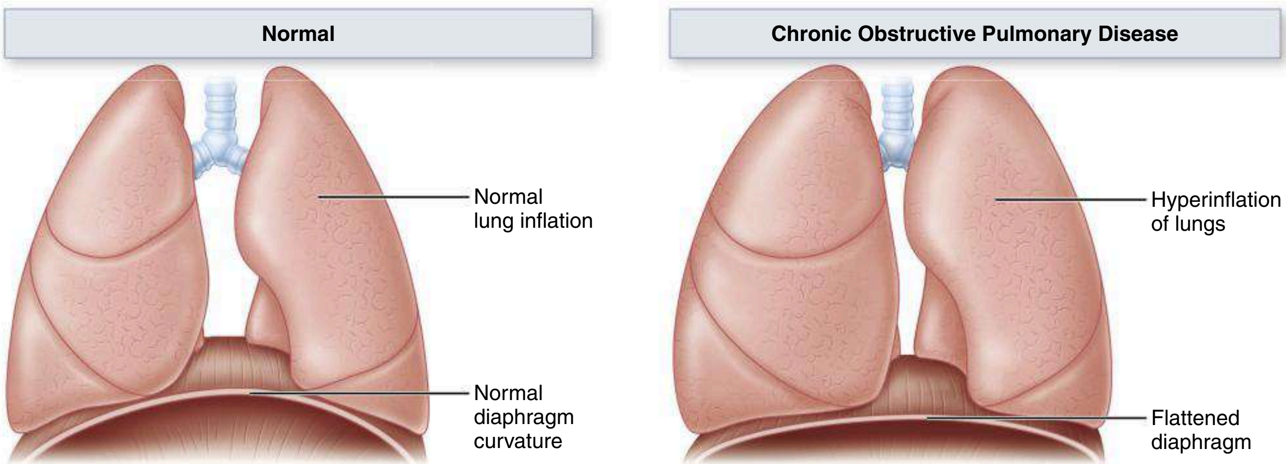

FIG. 51.5 Diaphragm shape and lung inflation in the normal client and in the client with chronic obstructive pulmonary disease.

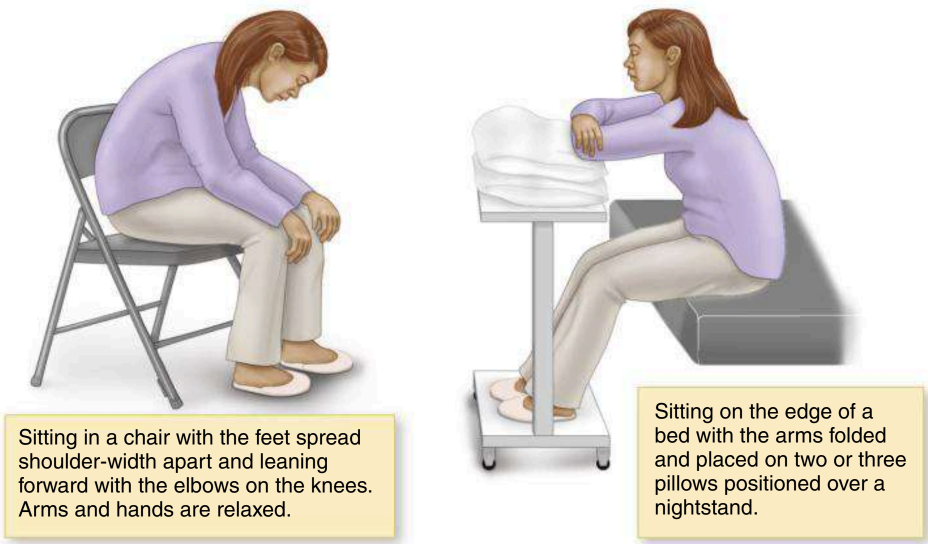

FIG. 51.6 Orthopnea positions that clients with chronic obstructive pulmonary disease can assume to ease the work of breathing.

BOX 51.6 Client Education: Chronic Obstructive Pulmonary Disease

- Adhere to activity limitations, alternating rest periods with activity.
- Avoid eating gas-producing foods, spicy foods, and extremely hot or cold foods.
- Avoid exposure to individuals with infections, and avoid crowds.
- Avoid extremes in temperature.
- Avoid fireplaces, pets, feather pillows, and other environmental allergens.
- Avoid powerful odors.
- Meet nutritional requirements.
- Receive immunizations as recommended.
- Recognize the signs and symptoms of respiratory infection and hypoxia.
- Stop smoking.
- Use medications and inhalers and/or nebulizers as prescribed.
- Use oxygen therapy as prescribed.
- Use pursed-lip and diaphragmatic or abdominal breathing.
- When dusting, use a wet cloth.

5. A sputum culture identifies the organism.
6. The white blood cell count, procalcitonin, and the erythrocyte sedimentation rate are elevated.

B. Assessment
1. Chills
2. Elevated temperature, other vital sign abnormalities
3. Pleuritic pain
4. Myalgia
5. Tachypnea, tachycardia
6. Rhonchi and wheezes
7. Use of accessory muscles for breathing, dyspnea
8. Cyanosis, especially around the mouth or conjunctiva
9. Mental status changes
10. Sputum production, hemoptysis

C. Interventions
1. Administer oxygen as prescribed.
2. Monitor respiratory status.
3. Monitor for labored respirations, cyanosis, and cold and clammy skin.
4. Encourage coughing and deep breathing and use of the incentive spirometer.
5. Place the client in a semi-Fowler's position to facilitate breathing and lung expansion.
6. Change the client's position frequently and ambulate as tolerated to mobilize secretions.
7. Provide CPT (see Chapter 70).
8. Perform nasotracheal suctioning if the client is unable to clear secretions.
9. Monitor pulse oximetry.
10. Monitor and record color, consistency, and amount of sputum.
11. Provide a high-calorie, high-protein diet with small frequent meals.
12. Monitor intake and output.
13. Encourage fluids, up to 3 L/day, to thin secretions unless contraindicated.
14. Provide a balance of rest and activity, increasing activity gradually.
15. Administer antibiotics as prescribed.
16. Administer antipyretics, bronchodilators, cough suppressants, mucolytic agents, and expectorants as prescribed.
17. Prevent the spread of infection by handwashing and the proper disposal of secretions.

D. Client education
1. About the importance of rest, proper nutrition, and adequate fluid intake

2. To avoid chilling and exposure to individuals with respiratory infections or viruses
3. To avoid smoke exposure
4. Regarding medications and the use of inhalants as prescribed
5. To notify the PHCP if chills, fever, dyspnea, hemoptysis, or increased fatigue occurs
6. To receive a pneumococcal vaccine as recommended by the PHCP; refer to Chapter 19 and the following website for information about this vaccine: http://www.cdc.gov/vaccines/vpd-vac/pneumo/default.htm

Teach clients that using proper handwashing techniques, disposing of respiratory secretions properly, and receiving vaccines will assist in preventing the spread of infection.

VIII. Severe Acute Respiratory Syndrome (SARS)

A. Respiratory illness caused by the coronavirus called *SARS-associated coronavirus*
B. The syndrome begins with a fever, an overall feeling of discomfort, body aches, and mild respiratory symptoms.
C. After 2 to 7 days, the client may develop a dry cough and dyspnea.
D. Infection is spread by close person-to-person contact or by direct contact with infectious material (respiratory secretions from infected persons or contact with objects contaminated with infectious droplets).
E. Prevention includes avoiding contact with those suspected of having SARS, avoiding travel to countries where an outbreak of SARS exists, avoiding close contact with crowds in areas where SARS exists, and frequent handwashing if in an area where SARS exists.

IX. COVID-19 (Coronavirus)

A. Description
1. SARS-CoV-2 is the coronavirus that causes COVID-19.
2. Older adults and people who have severe underlying medical conditions such as heart or lung disease or diabetes are at higher risk for developing more serious complications from COVID-19 illness.

B. Symptoms
1. People with COVID-19 have had a wide range of symptoms reported, ranging from mild symptoms to severe illness. Symptoms may appear 2 to 14 days after exposure to the virus. Symptoms can include the following:
   a. Fever or chills
   b. Cough
   c. Shortness of breath or difficulty breathing
   d. Fatigue
   e. Muscle or body aches
   f. Headache
   g. New loss of taste or smell
   h. Sore throat
   i. Congestion or runny nose
   j. Nausea or vomiting
   k. Diarrhea
2. Emergency care should be sought if the person is having difficulty breathing, experiences persistent pain or pressure in the chest, if the person experiences new confusion, is unable to stay awake, or if cyanosis develops.

C. Transmission and prevention
1. COVID-19 may be spread by people who are not showing symptoms.
2. Transmission is from person to person and via contact with the virus via respiratory droplets produced when an infected person coughs, sneezes, or talks.
3. These droplets can land in the mouths or noses of people who are nearby or possibly be inhaled into the lungs.
4. Droplet precautions and possibly contact precautions are necessary.
5. Prevention includes avoiding crowds; maintaining 6 feet of social distancing, especially from sick people; handwashing; the use of hand sanitizer; wearing masks; coughing and sneezing into the elbow; keeping the hands away from the face; and keeping frequently touched surfaces cleaned and sanitized daily.
6. Prophylactic treatment may be recommended for those exposed to coronavirus and may include selected vitamins and minerals, antiviral medication, and immune system booster supplements; the person should not begin any prophylactic treatment unless recommended by the primary health care provider.
7. All eligible individuals are highly encouraged to receive the coronavirus vaccine and recommended boosters to protect self and others from contracting the virus.

D. Treatment
1. Varies depending on the clinical presentation
2. The person needs to report symptoms immediately if coronavirus is suspected and seek treatment from the primary health care provider.
3. The primary health care provider will prescribe treatment based on the most current CDC guidelines for treating coronavirus.
4. For additional information, refer to the Centers for Disease Control and Prevention: Coronavirus 2019 (COVID-19): https://www.cdc.gov/coronavirus/2019-ncov/faq.html

X. Influenza

A. Description
1. Also known as the flu; highly contagious acute viral respiratory infection
2. May be caused by several viruses, usually known as types A, B, and C
3. Yearly vaccination is recommended to prevent the disease, especially for those older than 50 years of age, individuals with chronic illness or who are immunocompromised, those living in institutions, and health care personnel providing direct care to clients (the vaccination is contraindicated in the individual with egg allergies).
4. Additional prevention measures include avoiding those who have developed influenza, frequent and proper handwashing, and cleaning and disinfecting surfaces that have become contaminated with secretions.
5. Avian influenza A (H5N1)
   a. Affects birds; does not usually affect humans; however, human cases have been reported in some countries.
   b. An H5N1 vaccine has been developed for use if a pandemic virus were to emerge.
   c. Reported symptoms are similar to those associated with influenza types A, B, and C.
   d. Prevention measures include thorough cooking of poultry products, avoiding contact with wild animals, frequent and proper handwashing, and cleaning and disinfecting surfaces that have become contaminated with secretions.
6. Swine (H1N1) influenza
   a. A strain of flu that consists of genetic materials from swine, avian, and human influenza viruses
   b. Signs and symptoms are similar to those that present with seasonal flu; in addition, vomiting and diarrhea commonly occur.
   c. Prevention measures and treatment are the same as for the seasonal flu.

B. Refer to Chapter 19 for information on vaccines.

C. Assessment
1. Acute onset of fever, chills, and muscle aches
2. Headache
3. Fatigue, weakness, anorexia
4. Sore throat, cough, and rhinorrhea

D. Interventions
1. Encourage rest.
2. Encourage fluids to prevent pulmonary complications (unless contraindicated).
3. Monitor lung sounds.
4. Provide supportive therapy such as antipyretics or antitussives as indicated.
5. Administer antiviral medications as prescribed for the current strain of influenza (see Chapter 52).

E. Client education
1. Thoroughly wash hands especially after blowing nose, sneezing, coughing, rubbing eyes, or touching face.
2. Avoid crowded places; stay home if not feeling well.
3. Cover mouth with tissue when sneezing or coughing.
4. Cough or sneeze in the upper sleeve rather than in the hand.

XI. Legionnaire's Disease

A. Description
1. Acute bacterial infection caused by *Legionella pneumophila*
2. Sources of the organism include contaminated cooling tower water and warm stagnant water supplies, including water vaporizers, water sonicators, whirlpool spas, and showers.
3. Person-to-person contact does not occur; the risk for infection is increased by the presence of other conditions.

B. Assessment: Influenza-like symptoms with a high fever, chills, muscle aches, and headache that may progress to dry cough, pleurisy, and sometimes diarrhea

C. Interventions: Treatment is supportive, and antibiotics may be prescribed.

XII. Pleural Effusion

A. Description
1. Pleural effusion is the collection of fluid in the pleural space.
2. Any condition that interferes with secretion or drainage of this fluid will lead to pleural effusion.

B. Assessment
1. Pleuritic pain that is sharp and increases with inspiration
2. Progressive dyspnea with decreased movement of the chest wall on the affected side
3. Dry, nonproductive cough caused by bronchial irritation
4. Tachycardia
5. Elevated temperature
6. Decreased breath sounds and/or pleural friction rub over affected area/side
7. Chest x-ray film or CT scan that shows pleural effusion and a mediastinal shift away from the fluid if the effusion is more than 250 mL

C. Interventions
1. Identify and treat the underlying cause.
2. Monitor breath sounds.
3. Place the client in a Fowler's position.
4. Encourage coughing and deep breathing.
5. Prepare the client for thoracentesis.
6. If pleural effusion is recurrent, prepare the client for pleurectomy or pleurodesis as prescribed.

D. Pleurectomy
   1. Consists of surgically stripping the parietal pleura away from the visceral pleura
   2. This produces an intense inflammatory reaction that promotes adhesion formation between the two layers during healing.

E. Pleurodesis
   1. Involves the instillation of a sclerosing substance into the pleural space via a thoracotomy tube
   2. The substance creates an inflammatory response that scleroses tissue together.

XIII. Empyema

A. Description
   1. Collection of pus within the pleural cavity
   2. The fluid is thick, opaque, and foul-smelling.
   3. The most common cause is pulmonary infection and lung abscess caused by thoracic surgery or chest trauma, in which bacteria are introduced directly into the pleural space.
   4. Treatment focuses on treating the infection, emptying the empyema cavity, reexpanding the lung, and controlling the infection.

B. Assessment
   1. Recent febrile illness or trauma
   2. Chest pain
   3. Cough
   4. Dyspnea
   5. Anorexia and weight loss
   6. Malaise
   7. Elevated temperature and chills
   8. Night sweats
   9. Pleural exudate on chest x-ray

C. Interventions
   1. Monitor breath sounds.
   2. Place the client in a semi-Fowler's or high-Fowler's position.
   3. Encourage coughing and deep breathing.
   4. Administer antibiotics as prescribed.
   5. Instruct the client to splint the chest as necessary.
   6. Assist with thoracentesis or chest tube insertion to promote drainage and lung expansion.
   7. If marked pleural thickening occurs, prepare the client for decortication, if prescribed; this surgical procedure involves removal of the restrictive mass of fibrin and inflammatory cells.

XIV. Pleurisy

A. Description
   1. Inflammation of the visceral and parietal membranes; may be caused by pulmonary infarction or pneumonia
   2. The visceral and parietal membranes rub together during respiration and cause pain.
   3. Pleurisy usually occurs on one side of the chest, usually in the lower lateral portions in the chest wall.

B. Assessment
   1. Knifelike pain aggravated on deep breathing and dry cough
   2. Dyspnea
   3. Pleural friction rub heard on auscultation

C. Interventions
   1. Identify and treat the cause.
   2. Monitor lung sounds.
   3. Administer analgesics as prescribed.
   4. Apply hot or cold applications as prescribed.
   5. Encourage coughing and deep breathing.
   6. Instruct the client to lie on the affected side to splint the chest.

XV. Pulmonary Embolism

A. Description
   1. Occurs when a thrombus forms (most commonly in a deep vein), detaches, travels to the right side of the heart, and then lodges in a branch of the pulmonary artery
   2. May be classified as massive, submassive, or low-risk pulmonary embolism
   3. Clients prone to pulmonary embolism are those at risk for deep vein thrombosis, including those with prolonged immobilization, surgery, obesity, pregnancy, heart failure, advanced age, a history of thromboembolism, or a client who takes an estrogen-containing therapy.
   4. Fat emboli can occur as a complication following fracture of a long bone and can cause pulmonary emboli.
   5. Treatment is aimed at prevention through risk factor recognition and elimination.

B. Assessment (Box 51.7)

C. Interventions (see Clinical Judgment: Take Action Box)

BOX 51.7 Assessment Findings: Pulmonary Embolism

- Apprehension and restlessness
- Blood-tinged sputum, hemoptysis
- Chest pain
- Cough
- Crackles and wheezes on auscultation
- Cyanosis
- Diaphoresis
- Distended neck veins
- Dyspnea accompanied by anginal and pleuritic pain, exacerbated by inspiration
- Feeling of impending doom
- Hypotension
- Petechiae over the chest and axilla
- Shallow respirations, increased respiratory rate
- Tachypnea and tachycardia

CLINICAL JUDGMENT: TAKE ACTION

The nurse is caring for a client 2 days postoperative. The client has a history of deep vein thrombosis and heart failure. The client also has difficulty with mobility and is obese. The client calls the nurse and reports sudden chest pain, a cough, and difficulty breathing. The client is anxious and restless. Respirations are 26 breaths per minute and shallow. Pulse is 120 beats per minute. Blood pressure is 90/66 mm Hg. Pulse oximetry reading is 89%. Crackles are heard on auscultation of the lungs. The nurse suspects the development of pulmonary embolism and takes the following actions:

- Reassures the client and elevates the head of the bed
- Notifies the Rapid Response Team and the PHCP
- Administers oxygen
- Prepares to obtain an arterial blood gas
- Prepares for laboratory studies to be drawn and diagnostic scanning
- Prepares for the administration of heparin therapy or other therapies
- Monitors vital signs and checks lung sounds
- Provides comfort and emotional support
- Documents the event, interventions taken, and the client's response to treatment

XVI. Lung Cancer and Laryngeal Cancer: See Chapter 45 for more information.

XVII. Carbon Monoxide Poisoning: See Chapter 43 for more information.

XVIII. Histoplasmosis

A. Description
1. Pulmonary fungal infection caused by spores of *Histoplasma capsulatum*
2. Transmission occurs by the inhalation of spores, which commonly are found in contaminated soil.
3. Spores also are usually found in bird droppings.

B. Assessment
1. Similar to pneumonia
2. Positive skin test for histoplasmosis
3. Positive agglutination test
4. Splenomegaly, hepatomegaly

C. Interventions
1. Administer oxygen as prescribed.
2. Monitor breath sounds.
3. Administer antiemetics, antihistamines, antipyretics, and corticosteroids as prescribed.
4. Administer fungicidal medications as prescribed.
5. Encourage coughing and deep breathing.
6. Place the client in a semi-Fowler's position.
7. Monitor vital signs.
8. Monitor for nephrotoxicity from fungicidal medications.
9. Instruct the client to wear a mask and spray the floor with water before sweeping a barn or chicken coop.

XIX. Sarcoidosis

A. Description
1. Presence of epithelioid cell tubercles in the lung
2. The cause is unknown, but a high titer of Epstein-Barr virus may be noted.

B. Assessment
1. Night sweats
2. Chest pain
3. Fever
4. Weight loss
5. Cough and dyspnea
6. Skin nodules
7. Polyarthritis
8. Kveim test: Sarcoid node antigen is injected intradermally and causes a local nodular lesion in about 1 month.

C. Interventions
1. Administer corticosteroids to control symptoms.
2. Monitor temperature.
3. Increase fluid intake.
4. Provide frequent periods of rest.
5. Encourage small, frequent, nutritious meals.

XX. Occupational Lung Disease

A. Description
1. Caused by exposure to environmental or occupational fumes, dust, vapors, gases, bacterial or fungal antigens, and allergens; can result in acute reversible effects or chronic lung disease
2. Common disease classifications include occupational asthma pneumoconiosis (silicosis or coal miner's [black lung] disease), diffuse interstitial fibrosis (asbestosis, talcosis, berylliosis), or extrinsic allergic alveolitis (farmer's lung, bird fancier's lung, or machine operator's lung).

B. Assessment: Manifestations depend on the type of disease and respiratory symptoms.

C. Interventions
1. Prevention through the use of respiratory protective devices
2. Treatment is based on the symptoms experienced by the client.

XXI. Tuberculosis

A. Description
1. Highly communicable disease caused by *Mycobacterium tuberculosis*
2. *M. tuberculosis* is a nonmotile, nonsporulating, acid-fast rod that secretes niacin; when the bacillus reaches a susceptible site, it multiplies freely.

3. Because *M. tuberculosis* is an aerobic bacterium, it primarily affects the pulmonary system, especially the upper lobes, where the oxygen content is highest, but it also can affect other areas of the body, such as the brain, intestines, peritoneum, kidney, joints, and liver.
4. An exudative response causes a nonspecific pneumonitis and the development of granulomas in the lung tissue.
5. Tuberculosis has an insidious onset, and many clients are not aware of symptoms until the disease is well advanced.
6. Improper or noncompliant use of treatment programs may cause the development of mutations in the tubercle bacilli, resulting in a **multidrug-resistant** strain of **tuberculosis** (MDR-TB).
7. The goal of treatment is to prevent transmission, control symptoms, and prevent progression of the disease.

B. Risk factors (Box 51.8)

BOX 51.8 Risk Factors for Tuberculosis

- Child younger than 5 years of age
- Drinking unpasteurized milk if the cow is infected with bovine tuberculosis
- Homeless individuals or those from a lower socioeconomic group, minority group, or refugee group
- Individuals in constant, frequent contact with an untreated or undiagnosed individual
- Individuals living in crowded areas, such as long-term care facilities, prisons, and mental health facilities
- Older client
- Individuals with malnutrition, infection, immune dysfunction, or human immunodeficiency virus infection; or immunosuppressed as a result of medication therapy
- Individuals who abuse alcohol or are intravenous drug users
- History of past exposure, travel to other countries

C. Transmission
1. Via the airborne route by droplet infection
2. When an infected individual coughs, laughs, sneezes, or sings, droplet nuclei containing tuberculosis bacteria enter the air and may be inhaled by others.
3. Identification of those in close contact with the infected individual is important so that they can be tested and treated as necessary.
4. When contacts have been identified, these persons are assessed with a **tuberculin skin test** and chest x-rays to determine infection with tuberculosis.
5. After the infected individual has received tuberculosis medication for 2 to 3 weeks, the risk of transmission is reduced greatly.

D. Disease progression
1. Droplets enter the lungs, and the bacteria form a tubercle lesion.
2. The defense systems of the body encapsulate the tubercle, leaving a scar.
3. If encapsulation does not occur, bacteria may enter the lymph system, travel to the lymph nodes, and cause an inflammatory response termed *granulomatous inflammation*.
4. Primary lesions form; the primary lesions may become dormant but can be reactivated and become a secondary infection when reexposed to the bacterium.
5. In an active phase, tuberculosis can cause necrosis and cavitation in the lesions, leading to rupture, the spread of necrotic tissue, and damage to various parts of the body.

E. Client history
1. Past exposure to tuberculosis
2. Client's country of origin and travel to foreign countries in which the incidence of tuberculosis is high
3. Recent history of influenza, pneumonia, febrile illness, cough, or foul-smelling sputum production
4. Previous tests for tuberculosis; results of the testing
5. Recent **bacillus Calmette-Guérin (BCG) vaccine** (a vaccine containing attenuated tubercle bacilli that may be given to persons in foreign countries or to persons traveling to foreign countries to produce increased resistance to tuberculosis)

An individual who has received a BCG vaccine will have a positive tuberculin skin test result and should be evaluated for tuberculosis with a chest x-ray.

F. Clinical manifestations
1. May be asymptomatic in primary infection
2. Fatigue
3. Lethargy
4. Anorexia
5. Weight loss
6. Low-grade fever
7. Chills
8. Night sweats
9. Persistent cough lasting up to 3 weeks and the production of mucoid and mucopurulent sputum, which is occasionally streaked with blood
10. Chest tightness and a dull, aching chest pain may accompany the cough.

G. Chest assessment
1. A physical examination of the chest does not provide conclusive evidence of tuberculosis.
2. A chest x-ray is not definitive, but the presence of multinodular infiltrates with calcification in the upper lobes suggests tuberculosis.
3. If the disease is active, caseation and inflammation may be seen on the chest x-ray.
4. Advanced disease

TABLE 51.1 Classification of the Tuberculin Skin Test Reaction

Induration ≥5 mm Considered Positive in:	Induration ≥10 mm Considered Positive in:	Induration ≥15 mm Considered Positive in:
HIV-infected persons	Recent immigrants from high-prevalence countries	Any person, including persons with no known risk factors for TB
Recent contact of a person with TB disease	Injection drug users	
Persons with fibrotic changes on chest x-ray consistent with prior TB	Residents and employees in high-risk congregate settings	
Clients with organ transplants	Mycobacteriology laboratory personnel	
Persons immunosuppressed for other reasons	Persons with clinical conditions that place them at high risk	
	Children <4 years of age	
	Infants, children, and adolescents exposed to adults in high-risk categories	

HIV, Human immunodeficiency virus; *TB*, tuberculosis.
From Centers for Disease Control and Prevention: *Tuberculosis (TB) fact sheets* (website): https://www.cdc.gov/tb/publications/factsheets/testing/skintesting.htm

a. Dullness with **percussion** over involved parenchymal areas, bronchial breath sounds, rhonchi, and crackles indicate advanced disease.
b. Partial obstruction of a bronchus caused by endobronchial disease or compression by lymph nodes may produce localized **wheezing** and dyspnea.

H. QuantiFERON-TB Gold test
1. A blood analysis test by an enzyme-linked immunosorbent assay
2. A sensitive and rapid test (results can be available in 24 hours) that assists in diagnosing the client

I. Sputum cultures
1. Sputum specimens are obtained for an acid-fast smear.
2. A sputum culture identifying *M. tuberculosis* confirms the diagnosis.
3. May take up to 6 weeks to become positive.
4. After medications are started, sputum samples are obtained again to determine the effectiveness of therapy.
5. Most clients have negative cultures after 3 months of treatment.

J. **Tuberculin skin test (TST)** (Table 51.1)
1. A positive reaction does not mean that active disease is present but indicates previous exposure to tuberculosis or the presence of inactive (dormant) disease.
2. Once the test result is positive, it will be positive in any future tests.
3. Skin test interpretation depends on two factors: measurement in millimeters of the induration and the person's risk of being infected with tuberculosis and progression to disease if infected.
4. Once an individual's skin test is positive, a chest x-ray is necessary to rule out active tuberculosis or to detect old healed lesions.

K. The hospitalized client
1. The client with active tuberculosis is placed under airborne isolation precautions in a negative-pressure room; to maintain negative pressure, the door of the room must be tightly closed.
2. The room should have at least six exchanges of fresh air per hour and should be ventilated to the outside environment, if possible.
3. The nurse wears a particulate respirator (a special individually fitted mask) when caring for the client and a gown when the possibility of clothing contamination exists.
4. Thorough handwashing is required before and after caring for the client.
5. If the client needs to leave the room for a test or procedure, the client is required to wear a surgical mask.
6. Respiratory isolation is discontinued when the client is no longer considered infectious. An N95 mask should be worn when entering the room until isolation is discontinued.
7. After the infected individual has received tuberculosis medication for 2 to 3 weeks, the risk of transmission is reduced greatly.

L. Client education (Box 51.9)

BOX 51.9 Client Education: Tuberculosis

- Provide the client and family with information about tuberculosis and allay concerns about the contagious aspect of the infection.
- Instruct the client to follow the medication regimen exactly as prescribed and always to have a supply of the medication on hand.
- Advise the client that the medication regimen is continued up to 12 months, depending on the situation.
- Advise the client of the side and adverse effects of the medication and ways of minimizing them to ensure compliance.
- Reassure the client that after 2 to 3 weeks of medication therapy, it is unlikely that the client will infect anyone.
- Advise the client to resume activities gradually.
- Instruct the client about the need for adequate nutrition and a well-balanced diet (foods rich in iron, protein, and vitamin C) to promote healing and to prevent recurrence of the infection.
- Inform the client and family that respiratory isolation is not necessary because family members already have been exposed.
- Instruct the client to cover the mouth and nose when coughing or sneezing and to put used tissues into plastic bags.
- Instruct the client and family about thorough handwashing.
- Inform the client that a sputum culture is needed every 2 to 4 weeks once medication therapy is initiated.
- Inform the client that when the results of three sputum cultures are negative, the client is no longer considered infectious and usually can return to former employment.
- Advise the client to avoid excessive exposure to silicone or dust because these substances can cause further lung damage.
- Instruct the client regarding the importance of compliance with treatment, follow-up care, and sputum cultures, as prescribed.
- Notify the public health department.

PRACTICE QUESTIONS

1. The emergency department nurse is assessing a client who has sustained a blunt injury to the chest wall. Which finding indicates the presence of a pneumothorax in this client?
1. A low respiratory rate
2. Diminished breath sounds
3. The presence of a barrel chest
4. A sucking sound at the site of injury

2. The nurse is caring for a client hospitalized with acute exacerbation of chronic obstructive pulmonary disease. Which findings would the nurse expect to note on assessment of this client? **Select all that apply.**
- ❑ 1. A low arterial P_{CO_2} level
- ❑ 2. A hyperinflated chest noted on the chest x-ray
- ❑ 3. Decreased oxygen saturation with mild exercise
- ❑ 4. A widened diaphragm noted on the chest x-ray
- ❑ 5. Pulmonary function tests that demonstrate increased vital capacity

3. The nurse is preparing a list of home care instructions for a client who has been hospitalized and treated for tuberculosis. Which instructions would the nurse include on the list? **Select all that apply.**
- ❑ 1. Activities should be resumed gradually.
- ❑ 2. Avoid contact with other individuals, except family members, for at least 6 months.
- ❑ 3. A sputum culture is needed every 2 to 4 weeks once medication therapy is initiated.
- ❑ 4. Respiratory isolation is not necessary, because family members already have been exposed.
- ❑ 5. Cover the mouth and nose when coughing or sneezing and put used tissues in plastic bags.
- ❑ 6. When one sputum culture is negative, the client is no longer considered infectious and usually can return to former employment.

4. The nurse is caring for a client after a bronchoscopy and biopsy. Which finding, if noted in the client, would the nurse **immediately** report to the primary health care provider?
1. Dry cough
2. Hematuria
3. Bronchospasm
4. Blood-streaked sputum

5. The nurse is assessing the respiratory status of a client who has suffered a fractured rib. The nurse would expect to note which finding?
1. Slow, deep respirations
2. Rapid, deep respirations
3. Paradoxical respirations
4. Pain, especially with inspiration

6. A client with a chest injury has suffered flail chest. The nurse assesses the client for which **most** distinctive sign of flail chest?
1. Cyanosis
2. Hypotension
3. Paradoxical chest movement
4. Dyspnea, especially on exhalation

7. The nurse is assessing a client with multiple trauma who is at risk for developing acute respiratory distress syndrome. The nurse would assess for which **earliest** sign of acute respiratory distress syndrome?
1. Bilateral wheezing
2. Inspiratory crackles
3. Intercostal retractions
4. Increased respiratory rate

8. The nurse has conducted discharge teaching with a client diagnosed with tuberculosis who has been receiving medication for 2 weeks. The nurse determines that the client has understood the information if the client makes which statement?
1. "I need to continue medication therapy for 1 month."
2. "I can't shop at the mall for the next 6 months."
3. "I can return to work if a sputum culture comes back negative."
4. "I won't be contagious after 2 to 3 weeks of medication therapy."

9. The nurse is preparing to give a bed bath to an immobilized client with tuberculosis. The nurse would wear which items when performing this care?
1. Surgical mask and gloves
2. Particulate respirator, gown, and gloves
3. Particulate respirator and protective eyewear
4. Surgical mask, gown, and protective eyewear

10. A client has experienced pulmonary embolism. The nurse would assess for which symptom, which is **most** commonly reported?
1. Hot, flushed feeling
2. Sudden chills and fever
3. Chest pain that occurs suddenly
4. Dyspnea when deep breaths are taken

11. A client who is human immunodeficiency virus (HIV)–positive has had a tuberculin skin test (TST). The nurse notes a 7-mm area of induration at the site of the skin test and interprets the result as which finding?
1. Positive
2. Negative
3. Inconclusive
4. Need for repeat testing

12. A client with acquired immunodeficiency syndrome (AIDS) has histoplasmosis. The nurse would assess the client for which expected finding?
1. Dyspnea
2. Headache
3. Weight gain
4. Hypothermia

13. The nurse provides discharge instructions to a client with pulmonary sarcoidosis. The nurse concludes that the client understands the information if the client indicates to report which **early** sign of exacerbation?
1. Fever
2. Fatigue
3. Weight loss
4. Shortness of breath

14. The nurse is taking the history of a client with occupational lung disease (silicosis). The nurse would ask the client whether the client wears which item during periods of exposure to silica particles?
1. Mask
2. Gown
3. Gloves
4. Eye protection

15. The nurse is instructing a hospitalized client with a diagnosis of emphysema about measures that will enhance the effectiveness of breathing during dyspneic periods. Which position would the nurse instruct the client to assume?
1. Sitting up in bed
2. Side-lying in bed
3. Sitting in a recliner chair
4. Sitting up and leaning on an overbed table

16. The community health nurse is conducting an educational session with community members regarding the signs and symptoms associated with tuberculosis. The nurse informs the participants that tuberculosis is considered as a diagnosis if which signs and symptoms are present? **Select all that apply.**
❑ **1.** Dyspnea
❑ **2.** Headache
❑ **3.** Night sweats
❑ **4.** A bloody, productive cough
❑ **5.** A cough with the expectoration of mucoid sputum

17. The nurse performs an admission assessment on a client with a diagnosis of tuberculosis. The nurse would check the results of which diagnostic test that will confirm this diagnosis?
1. Chest x-ray
2. Bronchoscopy
3. Sputum culture
4. Tuberculin skin test

ANSWERS

1. *Answer:* 2
Rationale: This client has sustained a blunt or closed-chest injury. Basic symptoms of a closed pneumothorax are shortness of breath and chest pain. A larger pneumothorax may cause tachypnea, cyanosis, diminished breath sounds, and subcutaneous emphysema. Hyperresonance also may occur on the affected side. A sucking sound at the site of injury would be noted with an open chest injury.
Test-Taking Strategy: Focus on the **subject,** a blunt chest injury. Noting the word *blunt* will assist in eliminating option 4, which describes a sucking chest wound injury. Knowing that in a respiratory injury increased respirations will occur will assist you in eliminating option 1. Option 3 can be eliminated because a barrel chest is a characteristic finding in a client with chronic obstructive pulmonary disease.
References: Lewis, S., Harding, M., Kwong, J., Roberts, D., Hagler, D., & Reinisch, C. (2020). *Medical-surgical nursing: Assessment and management of clinical problems.* (11th ed.). St. Louis: Elsevier. p. 523; Urden, L., Stacy, K., & Lough, M. (2022). *Critical care nursing: Diagnosis and management.* (9th ed.). St. Louis: Elsevier. p. 447.

2. *Answer:* 2, 3
Rationale: Clinical manifestations of chronic obstructive pulmonary disease (COPD) include hypoxemia, hypercapnia, dyspnea on exertion and at rest, oxygen desaturation with exercise, and the use of accessory muscles of respiration. Chest x-rays reveal a hyperinflated chest and a flattened diaphragm if the disease is advanced. Pulmonary function tests will demonstrate decreased vital capacity.
Test-Taking Strategy: Focus on the **subject,** manifestations of COPD. Think about the pathophysiology associated with this disorder. Remember that hypercapnia, a hyperinflated chest, a flat diaphragm, oxygen desaturation on exercise, and decreased vital capacity are manifestations.
Reference: Lewis, S., Harding, M., Kwong, J., Roberts, D., Hagler, D., & Reinisch, C. (2020). *Medical-surgical nursing: Assessment and management of clinical problems.* (11th ed.). St. Louis: Elsevier. pp. 564-565.

3. *Answer:* 1, 3, 4, 5
Rationale: The nurse would provide the client and family with information about tuberculosis and allay concerns about the contagious aspect of the infection. The client needs to follow the medication regimen exactly as prescribed and always have a supply of the medication on hand. Side and adverse effects of the medication and ways of minimizing them to ensure compliance need to be explained. After 2 to 3 weeks of medication therapy, it is unlikely that the client will infect anyone. Activities need to be resumed gradually, and a well-balanced diet that is rich in iron, protein, and vitamin C to promote healing and prevent recurrence of infection needs to be consumed. Respiratory isolation is not necessary, because family members already have been exposed. Instruct the client about thorough handwashing, to cover the mouth and nose when coughing or sneezing, and to put used tissues into plastic bags. A sputum culture is needed every 2 to 4 weeks once medication therapy is initiated. When the results of three sputum cultures are negative, the client is no longer considered infectious and can usually return to former employment.
Test-Taking Strategy: Focus on the **subject,** home care instructions for tuberculosis. Knowledge regarding the pathophysiology, transmission, and treatment of tuberculosis is needed to answer this question. Read each option carefully to answer correctly.
Reference: Ignatavicius, D., Workman, M., Rebar, C., & Heimgartner, N. (2021). *Medical-surgical nursing: Concepts for interprofessional collaborative care.* (10th ed.). St. Louis: Elsevier. pp. 578, 580.

4. *Answer:* 3
Rationale: If a biopsy was performed during a bronchoscopy, blood-streaked sputum is expected for several hours. Frank blood indicates hemorrhage. A dry cough may be expected. The client needs to be assessed for signs of complications, which would include cyanosis, dyspnea, stridor, bronchospasm, hemoptysis, hypotension, tachycardia, and dysrhythmias. Hematuria is unrelated to this procedure.
Test-Taking Strategy: Note the **strategic word,** *immediately*. Eliminate option 2 first because it is unrelated to the procedure. Next, eliminate option 1, because a dry cough may be expected. Noting that a biopsy has been performed will assist in eliminating option 4, because blood-streaked sputum would be expected. Note that the correct option relates to the airway.
References: Pagana, K., Pagana, T., & Pagana, T.N. (2021). *Mosby's diagnostic and laboratory test reference.* (15th ed.). St. Louis: Elsevier. p. 188.

5. *Answer:* 4
Rationale: Rib fractures result from a blunt injury or a fall. Typical signs and symptoms include shallow respirations, splinting or guarding the chest protectively to minimize chest movement, pain and tenderness localized at the fracture site that is exacerbated by inspiration and palpation, and possible bruising at the fracture site. Paradoxical respirations are seen with flail chest.
Test-Taking Strategy: Focus on the **subject,** findings associated with a rib fracture. Noting that the client sustained an injury and focusing on the anatomical location of the injury will direct you to the correct option.
References: Lewis, S., Harding, M., Kwong, J., Roberts, D., Hagler, D., & Reinisch, C. (2020). *Medical-surgical nursing: Assessment and management of clinical problems.* (11th ed.). St. Louis: Elsevier. p. 522; Urden, L., Stacy, K., & Lough, M. (2022). *Critical care nursing: Diagnosis and management.* (9th ed.). St. Louis: Elsevier. p. 445.

6. *Answer:* 3
Rationale: Flail chest results from multiple rib fractures. This results in a "floating" section of ribs. Because this section is unattached to the rest of the bony rib cage, this segment results in paradoxical chest movement. This means that the force of inspiration pulls the fractured segment inward, while the rest of the chest expands. Similarly, during exhalation, the segment balloons outward while the rest of the chest moves inward. This is a characteristic sign of flail chest.

Test-Taking Strategy: Note the **strategic word,** *most.* Cyanosis and hypotension occur with many different disorders, so eliminate options 1 and 2 first. From the remaining options, choose paradoxical chest movement over dyspnea on exhalation by remembering that a flail chest has broken rib segments that move independently of the rest of the rib cage.
Reference: Ignatavicius, D., Workman, M., Rebar, C., & Heimgartner, N. (2021). *Medical-surgical nursing: Concepts for interprofessional collaborative care.* (10th ed.). St. Louis: Elsevier. p. 607.

7. *Answer:* 4
Rationale: The earliest detectable sign of acute respiratory distress syndrome is an increased respiratory rate, which can begin from 1 to 96 hours after the initial insult to the body. This is followed by increasing dyspnea, air hunger, retraction of accessory muscles, and cyanosis. Breath sounds may be clear or consist of fine inspiratory crackles or diffuse coarse crackles.
Test-Taking Strategy: Note the **strategic word,** *earliest*. Eliminate option 3 first, because intercostal retraction is a later sign of respiratory distress. Of the remaining options, recall that adventitious breath sounds (options 1 and 2) would occur later than an increased respiratory rate.
Reference: Lewis, S., Harding, M., Kwong, J., Roberts, D., Hagler, D., & Reinisch, C. (2020). *Medical-surgical nursing: Assessment and management of clinical problems.* (11th ed.). St. Louis: Elsevier. pp. 1591-1592.

8. *Answer:* 4
Rationale: The client is continued on medication therapy for up to 12 months, depending on the situation. The client generally is considered noncontagious after 2 to 3 weeks of medication therapy. The client is instructed to wear a mask if there will be exposure to crowds until the medication is effective in preventing transmission. The client is allowed to return to work when the results of three sputum cultures are negative.
Test-Taking Strategy: Focus on the **subject,** client understanding of medication therapy. Knowing that the medication therapy lasts for up to 12 months helps you eliminate option 1 first. Knowing that three sputum cultures must be negative helps you eliminate option 3 next. From the remaining options, recalling that the client is not contagious after 2 to 3 weeks of therapy will direct you to the correct option.
Reference: Ignatavicius, D., Workman, M., Rebar, C., & Heimgartner, N. (2021). *Medical-surgical nursing: Concepts for interprofessional collaborative care.* (10th ed.). St. Louis: Elsevier. p. 580.

9. *Answer:* 2
Rationale: The nurse who is in contact with a client with tuberculosis needs to wear an individually fitted particulate respirator. The nurse also would wear gloves as per standard precautions. The nurse wears a gown when the possibility exists that the clothing could become contaminated, such as when giving a bed bath.
Test-Taking Strategy: Focus on the **subject,** precautions when caring for the client with tuberculosis. Think about the nurse's task, a bed bath. Knowing that the nurse needs to wear a particulate respirator eliminates options 1 and 4. Knowledge of basic standard precautions directs you to the correct option.
Reference: Potter, P., Perry, A. G., Stockert, P. A., & Hall, A. M. (2021). *Fundamentals of nursing.* (10th ed.). St. Louis: Elsevier. p. 438.

10. *Answer:* 3
Rationale: The most common initial symptom in pulmonary embolism is chest pain that is sudden in onset. The next most commonly reported symptom is dyspnea, which is accompanied by an increased respiratory rate. Other typical symptoms of pulmonary embolism include apprehension and restlessness, tachycardia, cough, and cyanosis.
Test-Taking Strategy: Note the **strategic word,** *most.* Because pulmonary embolism does not result from an infectious process or an allergic reaction, eliminate options 1 and 2 first. To select between the correct option and option 4, look at them closely. Option 4 states dyspnea when deep breaths are taken. Although dyspnea commonly occurs with pulmonary embolism, dyspnea is not associated only with deep breathing. Therefore, eliminate option 4.
Reference: Ignatavicius, D., Workman, M., Rebar, C., & Heimgartner, N. (2021). *Medical-surgical nursing: Concepts for interprofessional collaborative care.* (10th ed.). St. Louis: Elsevier. p. 588.

11. *Answer:* 1
Rationale: The client with HIV infection is considered to have positive results on tuberculin skin testing with an area of induration larger than 5 mm. The client without HIV is positive with an induration larger than 10 mm. The client with HIV is immunosuppressed, making a smaller area of induration positive for this type of client. It is possible for the client infected with HIV to have false-negative readings because of the immunosuppression factor. Options 2, 3, and 4 are incorrect interpretations.
Test-Taking Strategy: Eliminate options 3 and 4 first because they are **comparable or alike.** From the remaining options, recalling that the client with HIV infection is immunosuppressed will assist in determining the interpretation of the area of induration.
Reference: Ignatavicius, D., Workman, M., Rebar, C., & Heimgartner, N. (2021). *Medical-surgical nursing: Concepts for interprofessional collaborative care.* (10th ed.). St. Louis: Elsevier. p. 577.

12. *Answer:* 1
Rationale: Histoplasmosis is an opportunistic fungal infection that can occur in the client with AIDS. The infection begins as a respiratory infection and can progress to disseminated infection. Typical signs and symptoms include fever, dyspnea, cough, and weight loss. Enlargement of the client's lymph nodes, liver, and spleen may occur as well.
Test-Taking Strategy: Focus on the **subject,** manifestations of histoplasmosis. Recalling that histoplasmosis is an infectious process will help you eliminate option 4. Because the client has AIDS and another infection, weight gain is an unlikely symptom and can be eliminated next. Knowing that histoplasmosis begins as a respiratory infection helps you choose dyspnea over headache as the correct option.

Reference: Ignatavicius, D., Workman, M., Rebar, C., & Heimgartner, N. (2021). *Medical-surgical nursing: Concepts for interprofessional collaborative care.* (10th ed.). St. Louis: Elsevier. p. 332.

13. *Answer:* 4

Rationale: Dry cough and dyspnea are typical early manifestations of pulmonary sarcoidosis. Later manifestations include night sweats, fever, weight loss, and skin nodules.

Test-Taking Strategy: Note the **strategic word,** *early*. Because sarcoidosis is a pulmonary problem, eliminate options 1 and 3 first. Select the correct option over option 2 because the shortness of breath (and impaired ventilation) appears first and would cause the fatigue as a secondary symptom.

Reference: Lewis, S., Harding, M., Kwong, J., Roberts, D., Hagler, D., & Reinisch, C. (2020). *Medical-surgical nursing: Assessment and management of clinical problems.* (11th ed.). St. Louis: Elsevier. p. 532.

14. *Answer:* 1

Rationale: Silicosis results from chronic, excessive inhalation of particles of free crystalline silica dust. The client needs to wear a mask to limit inhalation of this substance, which can cause restrictive lung disease after years of exposure. Options 2, 3, and 4 are not necessary.

Test-Taking Strategy: Focus on the **subject,** prevention of silicosis. Recalling that exposure to silica dust causes the illness and that the dust is inhaled into the respiratory tract will direct you to the correct option.

Reference: Lewis, S., Harding, M., Kwong, J., Roberts, D., Hagler, D., & Reinisch, C. (2020). *Medical-surgical nursing: Assessment and management of clinical problems.* (11th ed.). St. Louis: Elsevier. pp. 516-517.

15. *Answer:* 4

Rationale: Positions that will assist the client with emphysema with breathing include sitting up and leaning on an overbed table, sitting up and resting the elbows on the knees, and standing and leaning against the wall.

Test-Taking Strategy: Eliminate options 1 and 3 first because they are **comparable or alike.** Next, eliminate option 2 because this position will not enhance breathing.

Reference: Ignatavicius, D., Workman, M., Rebar, C., & Heimgartner, N. (2021). *Medical-surgical nursing: Concepts for interprofessional collaborative care.* (10th ed.). St. Louis: Elsevier. pp. 544-545.

16. *Answer:* 1, 3, 4, 5

Rationale: Tuberculosis should be considered for any client with a persistent cough with mucoid sputum production, weight loss, anorexia, night sweats, hemoptysis, shortness of breath, fever, or chills. The client's previous exposure to tuberculosis needs to also be assessed and correlated with the clinical manifestations.

Test-Taking Strategy: Note the **subject,** clinical manifestations of tuberculosis. Note that headache is not specifically associated with tuberculosis, is not respiratory in nature, and is not associated with an infection to assist in eliminating this option.

Reference: Lewis, S., Harding, M., Kwong, J., Roberts, D., Hagler, D., & Reinisch, C. (2020). *Medical-surgical nursing: Assessment and management of clinical problems.* (11th ed.). St. Louis: Elsevier. pp. 510-511.

17. *Answer:* 3

Rationale: Tuberculosis is definitively diagnosed through culture and isolation of *Mycobacterium tuberculosis*. A presumptive diagnosis is made based on a tuberculin skin test, a sputum smear that is positive for acid-fast bacteria, a chest x-ray, and histological evidence of granulomatous disease on biopsy.

Test-Taking Strategy: Focus on the **subject,** confirming the diagnosis of tuberculosis. Confirmation is made by identifying the bacteria, *M. tuberculosis*.

Reference: Ignatavicius, D., Workman, M., Rebar, C., & Heimgartner, N. (2021). *Medical-surgical nursing: Concepts for interprofessional collaborative care.* (10th ed.). St. Louis: Elsevier. p. 578.

Level of Cognitive Ability: Analyzing
Client Needs: Physiological Integrity
Integrated Process: Nursing Process—Assessment
Clinical Judgment/Cognitive Skill: Recognize Cues
Content Area: Adult Health: Respiratory
Health Problem: Adult Health: Respiratory: Chest Injuries
Priority Concepts: Gas Exchange; Perfusion

Level of Cognitive Ability: Analyzing
Client Needs: Physiological Integrity
Integrated Process: Nursing Process—Assessment
Clinical Judgment/Cognitive Skill: Recognize Cues
Content Area: Adult Health: Respiratory
Health Problem: Adult Health: Respiratory: Obstructive Pulmonary Disease
Priority Concepts: Gas Exchange; Perfusion

Level of Cognitive Ability: Analyzing
Client Needs: Safe and Effective Care Environment
Integrated Process: Teaching and Learning
Clinical Judgment/Cognitive Skill: Generate Solutions
Content Area: Adult Health: Respiratory
Health Problem: Adult Health: Respiratory: Tuberculosis
Priority Concepts: Patient Education; Infection

Level of Cognitive Ability: Analyzing
Client Needs: Physiological Integrity
Integrated Process: Nursing Process—Implementation
Clinical Judgment/Cognitive Skill: Take Action
Content Area: Complex Care: Emergency Situations/Management
Health Problem: N/A
Priority Concepts: Clinical Judgment; Gas Exchange

Level of Cognitive Ability: Analyzing
Client Needs: Physiological Integrity
Integrated Process: Nursing Process—Assessment
Clinical Judgment/Cognitive Skill: Recognize Cues
Content Area: Adult Health: Respiratory
Health Problem: Adult Health: Respiratory: Chest Injuries
Priority Concepts: Gas Exchange; Pain

Level of Cognitive Ability: Analyzing
Client Needs: Physiological Integrity
Integrated Process: Nursing Process—Assessment
Clinical Judgment/Cognitive Skill: Recognize Cues
Content Area: Adult Health: Respiratory
Health Problem: Adult Health: Respiratory: Chest Injuries
Priority Concepts: Gas Exchange; Pain

Level of Cognitive Ability: Analyzing
Client Needs: Physiological Integrity
Integrated Process: Nursing Process—Assessment
Clinical Judgment/Cognitive Skill: Recognize Cues
Content Area: Adult Health: Respiratory
Health Problem: Adult Health: Respiratory: Acute Respiratory Distress Syndrome/Failure
Priority Concepts: Gas Exchange; Perfusion

Level of Cognitive Ability: Evaluating
Client Needs: Physiological Integrity
Integrated Process: Teaching and Learning
Clinical Judgment/Cognitive Skill: Evaluate Outcomes
Content Area: Adult Health: Respiratory
Health Problem: Adult Health: Respiratory: Tuberculosis
Priority Concepts: Patient Education; Infection

Level of Cognitive Ability: Applying
Client Needs: Safe and Effective Care Environment
Integrated Process: Nursing Process—Implementation
Clinical Judgment/Cognitive Skill: Take Action
Content Area: Foundations of Care: Infection Control
Health Problem: Adult Health: Respiratory: Tuberculosis
Priority Concepts: Infection; Safety

Level of Cognitive Ability: Analyzing
Client Needs: Physiological Integrity
Integrated Process: Nursing Process—Assessment
Clinical Judgment/Cognitive Skill: Recognize Cues
Content Area: Adult Health: Respiratory
Health Problem: Adult Health: Respiratory: Pulmonary Embolism
Priority Concepts: Gas Exchange; Perfusion

Level of Cognitive Ability: Analyzing
Client Needs: Physiological Integrity
Integrated Process: Nursing Process—Analysis
Clinical Judgment/Cognitive Skill: Analyze Cues
Content Area: Adult Health: Respiratory
Health Problem: Adult Health: Respiratory: Tuberculosis
Priority Concepts: Evidence; Infection

Level of Cognitive Ability: Analyzing
Client Needs: Physiological Integrity
Integrated Process: Nursing Process—Assessment
Clinical Judgment/Cognitive Skill: Recognize Cues
Content Area: Adult Health: Respiratory
Health Problem: Adult Health: Respiratory: Viral, Bacterial, Fungal Infections
Priority Concepts: Clinical Judgment; Infection